EVERGREEN

An Architecture for a Self-Evolving Deep-Space Communications Probe

Staying in touch with Earth across deep time: optical links, century-scale power, onboard autonomy, self-repair, and graceful degradation

Abstract

We specify the 'ideal' deep-space communications probe: one engineered to **maintain a link with Earth for as long as physically possible** while it recedes, ages, and loses power. Four clocks run against it — the link budget falls as $1/R^2$, the radioisotope power decays exponentially, the hardware degrades, and the round-trip light time eventually makes Earth control impossible. The Evergreen architecture answers each: an **optical-primary, growable aperture** (gaining ~87 dB over radio for the same size); a century-scale power chain (Am-241 / compact fission plus deep hibernation); a **radiation-hard, self-repairing AI core** that becomes more autonomous precisely as the link weakens; and a **graceful-degradation ladder** from full science to priority summaries to beacon to archive. We derive the governing relations, quantify them in eight figures, and show that the deepest value of a long-lived link is not bulk data — most of that returns early — but *presence*: a maintained channel and a relay seed for whatever comes next.

What is real here

Most of this architecture is an **extrapolation of demonstrated engineering**, not new physics: Voyager has operated for ~49 years; NASA's DSOC laser terminal returned data optically from beyond Mars in 2024; radioisotope decay, deep-space autonomy, and near-Shannon coding are mature. The genuinely forward-looking elements — in-flight aperture growth, hardware self-repair, and true on-board machine 'evolution' — are flagged as aspirational in Section 12. Nothing here requires violating physics; it requires winning a long fight against entropy.

Companion to the Orchard Program: an Evergreen probe is the communication-and-survival design of a single Orchard node, and its relay behavior (Section 10) is how isolated probes become a network.

1. The Problem: Four Clocks Running Down

A relativistic ship solves transport for passengers; a long-lived probe must solve something harder — staying *reachable*. Four independent processes erode the link:

(i) Distance. Free-space loss grows as the square of range, so a fixed transmitter's data rate falls as $1/R^2$. **(ii) Power.** Radioisotope sources decay exponentially, halving every few decades. **(iii) Aging.** Radiation, thermal cycling, and micrometeoroids degrade electronics and optics. **(iv) Control horizon.** Round-trip light time grows linearly with distance until commanding the probe in any useful cadence becomes impossible. The ideal probe is the one that fights all four at once, and the design choices below follow directly from that.

2. Design Principles

The link is a partnership. Half the budget lives on Earth; growing the ground collector is as powerful as anything on the probe. **Optical-first.** Photons buy enormous gain per gram. **Autonomy scales with distance.** The probe must run itself once Earth is hours, then years, away. **Degrade gracefully.** Never fail hard; shed capability in defined modes. **Be evolvable and relay-aware.** Reconfigure, self-repair, and — eventually — seed successors that extend the chain (Figure 1).

Figure 1 — The Evergreen probe: an AI core orchestrating evolvable subsystems

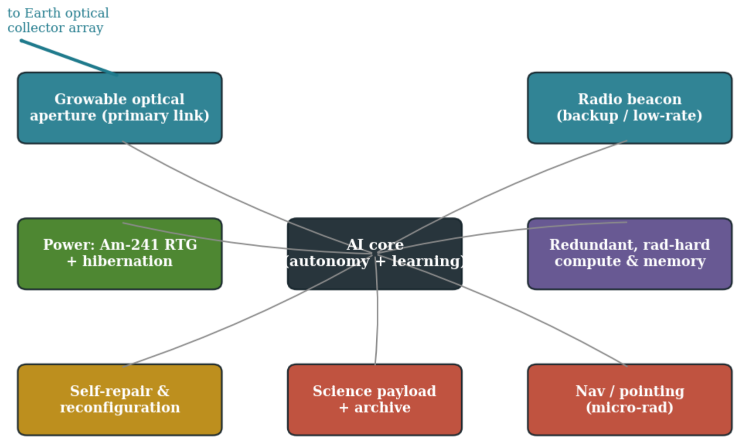

Figure 1. The Evergreen architecture: an AI core orchestrates an optical primary link, a radio beacon, century-scale power, redundant rad-hard compute, precision pointing, science/archive, and self-repair — all aimed at one goal, keeping the Earth channel alive.

3. The Link Budget and the 1/R^2 Wall

Received power follows the Friis relation, and the achievable data rate is that power divided by the noise per bit:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2}, \qquad R_b = \frac{P_r}{(E_b/N_0)\,N_0} \;\propto\; \frac{1}{R^2}$$

with antenna gain $G = \eta(\pi D/\lambda)^2$. Two consequences dominate the whole design. First, rate falls as the inverse square of distance (Figure 2): a Voyager-class radio link of a few hundred bits per second at ~165 AU becomes ~10^{-4} bps at Proxima — one bit per few hours. Second, every term except R is a lever we can pull: transmit gain G_t, receive gain G_r, and the noise floor N_0. The ultimate ceiling is Shannon's, $C = B \log_2(1 + S/N)$; good codes already operate within ~1 dB of it, so the remaining gains come from *aperture* and *wavelength*, not cleverer modulation.

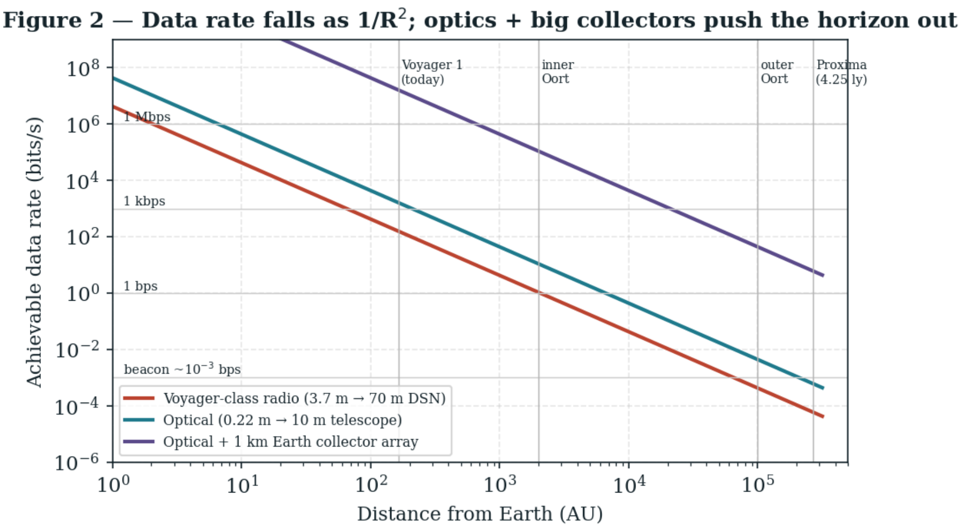

Figure 2. Data rate versus distance. Optical transmission and large Earth collectors shift the usable-link horizon outward by ~2 orders of magnitude; an optical link with a 1 km ground array still reaches Proxima at a few bits per second.

Architecture	Probe → Earth	Rate @165 AU	Rate @Proxima
Voyager-class radio	3.7 m → 70 m dish	~10^2–10^3 bps	~10^-4 bps
Optical, modest ground	0.22 m → 10 m scope	~10^3–10^4 bps	~10^-3 bps
Optical + km array	0.22 m → 1 km array	~10^6–10^7 bps	~few bps

Table 1. Idealized link comparison (near-Shannon coding assumed). The Earth-side aperture is the single biggest lever for an interstellar link.

4. Why Optical — and How to Keep Growing the Aperture

Because gain scales as $(D/\lambda)^2$, moving from X-band (3.6 cm) to a 1.55 μm laser shrinks the wavelength by ~23,000× and raises gain by its square:

$$\frac{G_{\mathrm{opt}}}{G_{\mathrm{radio}}} = \left(\frac{\lambda_{\mathrm{radio}}}{\lambda_{\mathrm{opt}}}\right)^2 \approx (2.3\times10^4)^2 \approx 5\times10^8 \ \ (\sim 87\ \mathrm{dB})$$

A 22 cm optical terminal thus out-gains a multi-metre dish (Figure 3), and optical receivers work at the quantum noise floor hν rather than the thermal floor kT. The costs are real: microradian pointing, photon-counting detectors, and sensitivity to the Sun-angle geometry. The ‘growable aperture’ principle addresses the $1/R^2$ wall from the probe side — deployable membrane optics or self-assembled reflectors that *enlarge* the effective aperture over the mission, trading stowed mass for end-of-life gain. Every doubling of aperture diameter is a factor of four in rate.

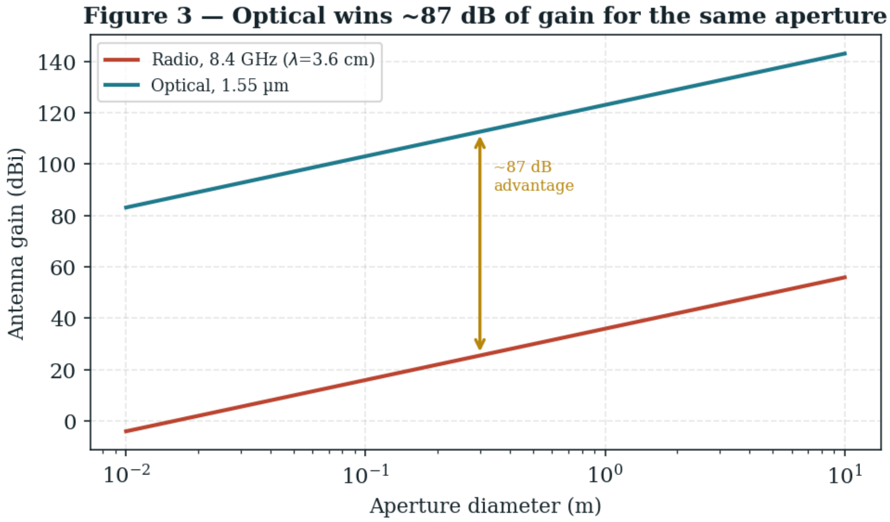

Figure 3. Antenna/telescope gain versus aperture for radio and optical bands. The ~87 dB optical advantage is why the ideal probe is optical-primary, with radio kept only as a low-rate beacon.

5. Power Across Deep Time

Power is the hardest deep-time constraint. A radioisotope source decays as

$$P(t) = P_0\left(\tfrac{1}{2}\right)^{t/t_{1/2}} \ \times\ (\text{thermocouple aging})$$

Plutonium-238 ($t_{1/2}$ = 87.7 yr) is the workhorse, but it halves within a human lifetime and Voyager already rations power down toward a hibernation floor. For century-scale missions the ideal probe uses a slower isotope — americium-241 ($t_{1/2}$ = 432 yr), now under development — or a compact fission source that stays near-constant until its fuel is spent (Figure 4, Table 2). Equally important is the demand side: aggressive duty-cycling and deep hibernation, waking only to transmit, can stretch a fixed energy budget across centuries.

Figure 4 — Power is the deep-time bottleneck: choose the isotope for the era

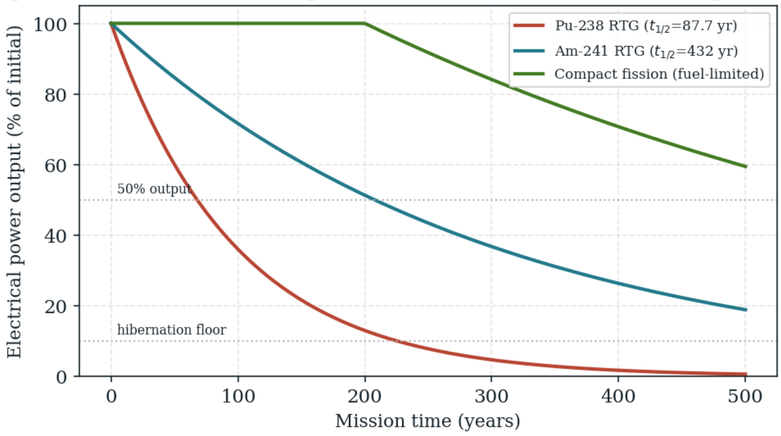

Figure 4. Electrical output versus time for three power chains. Choosing the isotope to the mission era — and hibernating between transmissions — is what keeps the transmitter alive at all.

Source	Half-life / lifetime	Trait
Pu-238 RTG	87.7 yr	Proven; halves within a lifetime
Am-241 RTG	432 yr	Lower power density, far better longevity
Compact fission	Fuel-limited (decades+)	Near-constant, but heavier / complex
Beamed / harvested	Source-dependent	Useful near stars; not for the deep dark

Table 2. Power-source options. The Evergreen baseline is Am-241 plus deep hibernation, with fission as a high-power variant.

6. The Autonomy Horizon

The round-trip light time sets a hard limit on how Earth can interact with the probe:

$$\tau_{\mathrm{rt}} = \frac{2R}{c}$$

At the Moon it is seconds; at 165 AU it is ~44 hours; at one light-year it is two years; at Proxima, 8.5 years (Figure 5). Beyond roughly a light-day, Earth cannot steer the probe in any meaningful cadence — the spacecraft must navigate, manage faults, schedule its own transmissions, and prioritize science *autonomously*. This is why the architecture is AI-centric: autonomy is not a luxury but the only thing that works once the probe crosses its control horizon (Table 3).

Figure 5 — The autonomy horizon: beyond it, the probe must run itself

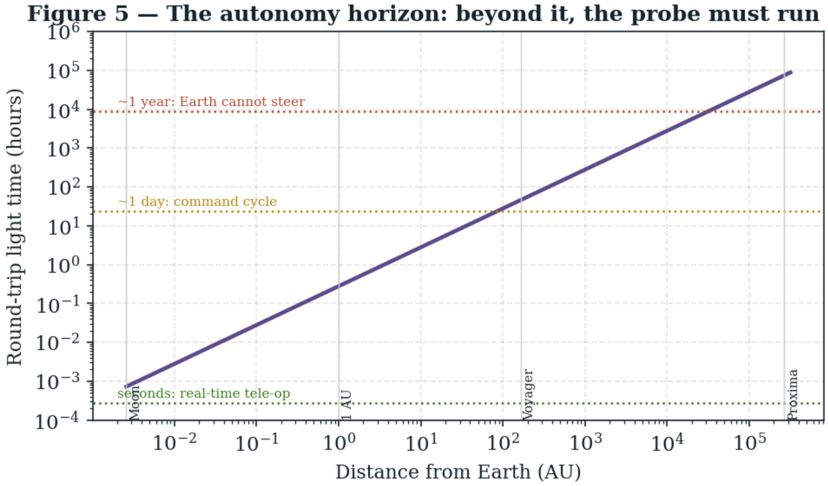

Figure 5. Round-trip light time versus distance. The probe passes from tele-operable, to daily-command, to fully self-governing as it recedes.

Regime	Round-trip light time	Required autonomy
Cislunar	seconds	Tele-operation possible
Inner Solar System	minutes–hours	Supervised autonomy, daily commands
Outer system / heliopause	hours–days	Goal-level tasking; local fault recovery
Interstellar (>0.01 ly)	months–years	Full self-government; Earth is an archive, not a pilot

Table 3. The autonomy horizon. Control authority migrates from Earth to the probe with distance.

7. The AI Core: What ‘Evolving’ Actually Means

‘AI-powered’ here is concrete, not decorative. The core performs five jobs that directly extend the link:

Adaptive link control — sensing the channel and choosing power, coding rate, modulation, and transmit timing (around solar conjunctions and noise) to maximize bits per joule. **Learned compression** — on-board models that distill raw observations into the few high-value bits worth the enormous cost of sending them. **Fault management** — detecting degradation and reconfiguring around it without waiting hours or years for Earth. **Science autonomy** — deciding what to observe and what to keep when it cannot ask. **Self-modeling** — maintaining a model of its own state to plan power, thermal, and lifetime budgets.

In the strong sense, ‘evolution’ means the probe’s behavior *improves with experience*: better anomaly detection, better compression tuned to what it actually sees, better link scheduling. Early in the mission Earth can still uplink improved software and models; past the autonomy horizon, the probe can only refine what it already carries. The honest boundary: adaptive control and learned compression are demonstrated technologies today; open-ended on-board self-improvement over centuries, on radiation-hardened hardware, is aspirational (Section 12).

8. Survival and Self-Repair

Over deep time a single component is unreliable. With a constant failure hazard λ, a unit’s survival to time t is $s(t)=e^{-\lambda t}$; with N independent redundant units the subsystem survives if at least one does:

$$s(t) = e^{-\lambda t}, \qquad S_{\mathrm{sub}} = 1 - (1 - s)^N$$

Redundancy alone is not enough across centuries; the ideal probe adds **active self-repair**: spare-swapping, software reconfiguration around dead memory, re-aiming and re-tuning, and — aspirationally — in-situ rebuilding of antennas or electronics from stored feedstock. The result is graceful decline rather than sudden death (Figure 6): capability erodes, but repair events and mode management keep the link alive far longer than passive hardware would.

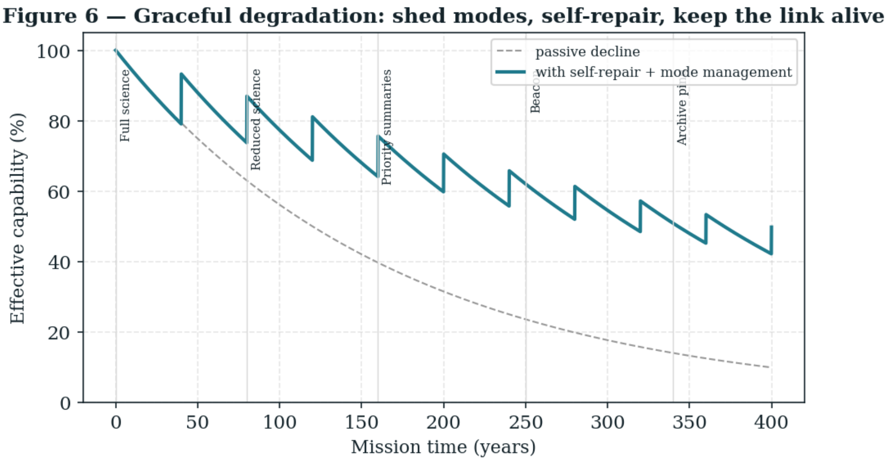

Figure 6. Effective capability versus time. Passive hardware (dashed) decays smoothly toward the floor; self-repair and disciplined mode management (solid) hold far more capability for far longer.

9. Graceful Degradation: From Streaming to Beacon to Archive

As rate and power fall, the probe descends a defined ladder of communication modes (Table 4), each cheaper than the last. Crucially, **most science data returns early** in the mission, while the probe is close and powered (Figure 7); deep-time communication is therefore less about volume than about *presence* — a maintained heartbeat, occasional high-value events, and a stored archive that can be dumped slowly or on request. This also exposes a real design tradeoff: a slower probe stays in good-link range far longer (a useful link for decades rather than years), at the cost of reaching its target later.

Mode	Trigger	What is sent
Full science	Close, fully powered	Raw / lightly compressed streams
Reduced science	Falling rate	Prioritized, heavily compressed products
Priority summaries	Marginal link	Distilled findings + state of health
Beacon	Sub-bps link	‘Alive + key parameters’ heartbeat, repeated
Archive	Power-critical	Silent store; dump slowly or on contact

Table 4. The communication-mode ladder. The AI selects the mode that maximizes value within the current power and link budget.

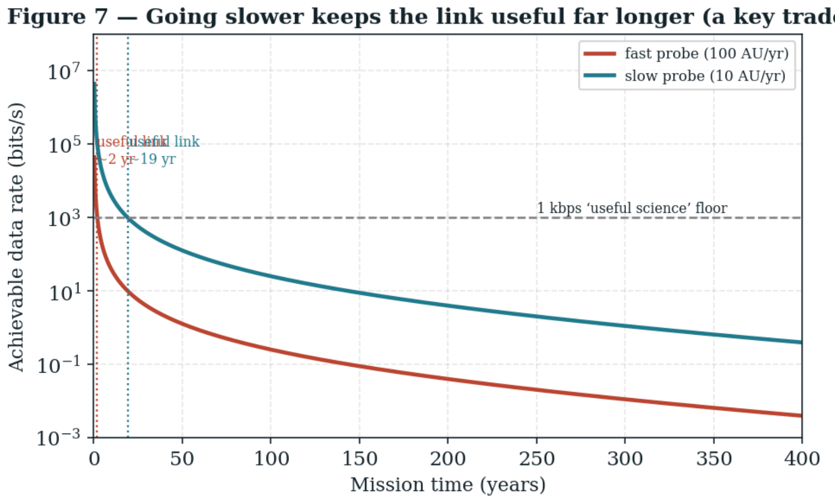

Figure 7. Achievable rate over mission time for a fast versus a slow probe. The slow probe keeps a useful (>1 kbps) link roughly ten times longer — the central tradeoff for a mission whose goal is staying in touch.

10. Relay-Aware Evolution: the Link as a Growing Network

A single probe's link must fall as $1/R^2$ — but a *chain* need not. If the probe can deploy or seed relay nodes behind it, each hop is short and strong, and the effective link degrades far more slowly than a direct shot. This is exactly the Orchard Program seed behavior: a probe that, late in life, manufactures or releases successor nodes turns a decaying point-to-point link into a self-extending network. The same optical terminals, autonomy, and power chain described here are what each relay needs. The ideal communicator, taken to its limit, is therefore not one probe but a **lineage** — each member an Evergreen, each planting the next.

11. The Evergreen Lifecycle

The mission is designed to age deliberately: it returns its bulk science early, hands control to itself at the autonomy horizon, grows aperture and self-repairs to defend the link, descends the mode ladder as power fades, and finally becomes an archival beacon and relay seed (Figure 8). It is built to become *more* independent precisely as it becomes fainter.

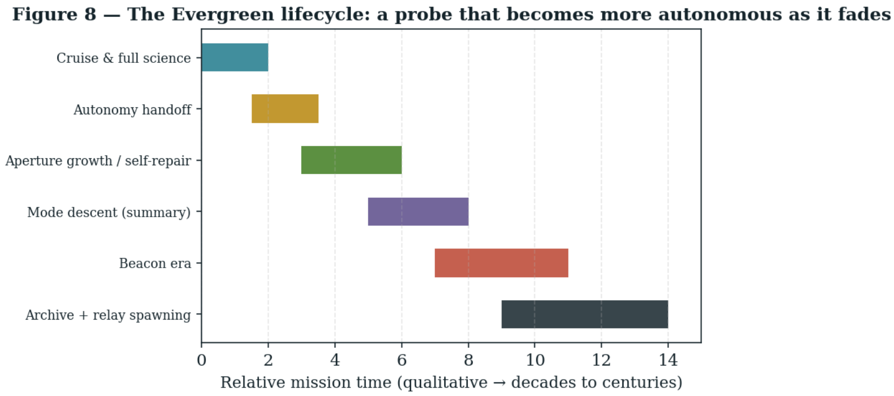

Figure 8. The Evergreen lifecycle. Capability migrates from Earth-directed science to autonomous presence and relay seeding over decades to centuries.

12. Demonstrated vs Aspirational

To keep the proposal honest, the load-bearing elements are sorted by maturity:

Element	Status
1/R^2 link physics; near-Shannon coding	Established
Optical deep-space comm (DSOC, 2024)	Demonstrated beyond Mars
Radioisotope power & decay; hibernation	Established (Voyager, RTGs)
Deep-space autonomy & fault management	Mature and improving
Am-241 RTG for century-scale power	In development
Adaptive link control + learned compression	Demonstrated in part
In-flight aperture growth (membrane / self-assembly)	Aspirational
Hardware self-repair / in-situ rebuilding	Aspirational
Open-ended on-board self-improvement over centuries	Aspirational
Relay-seed lineage (Orchard-style)	Speculative

Table 5. Maturity ledger. Unlike faster-than-light schemes, nothing here is forbidden by physics; the open problems are reliability, power, and autonomy over very long times.

13. Conclusion

The ideal AI-powered communications probe is the one that treats its own decline as a design input. It wins gain with optics and a growable aperture, buys time with a slow isotope and deep hibernation, governs itself once Earth falls below the horizon, defends the link by self-repair and disciplined mode descent, and — at the end — becomes a beacon and a seed rather than going dark. Its bulk science arrives early; its enduring gift is continuity. The link does not have to last forever to be worth building to last as long as possible.

“Built to become more independent precisely as it becomes fainter: the last thing it should do is not fall silent, but teach the next probe how to keep listening.”

Key Background & Benchmarks

Voyager mission longevity and X-band downlink (NASA/JPL). Deep Space Optical Communications (DSOC) on Psyche, optical downlink beyond Mars, 2024. Friis transmission and Shannon–Hartley capacity (standard). Radioisotope thermoelectric generators; Pu-238 ($t_{1/2}$=87.7 yr) and Am-241 ($t_{1/2}$=432 yr) heat sources. Deep Space Network 70 m and optical ground-station concepts. Reliability theory for redundancy and survival. Companion: the Orchard Program (self-replicating relay seeds and the 550 AU solar-lens layer). All numerical figures are order-of-magnitude engineering estimates, not quotations.